

eMudhra AAR Integration Documentation

<i>Document Name</i>	<i>eMudhra eSign Mobile App ASP Integration with AAR</i>
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Key Points

This document provides integration steps for using the eMudhra AAR file for eSign v2 transactions.

It outlines the process for initiating transactions, handling responses, invoking the eMudhra ARR Library, and processing results for OTP, Biometric, Face and Iris recognition modes.

Integration Process

01	02	03
Select or create PDF document	Authentication Mode Selection	Consent & perform eSign
<i>Prepare the document as per requirement for eSign v2 transaction</i>	<i>Choose between OTP, Biometric, Face or Iris recognition modes</i>	<i>User agrees to consent and completes the signing process</i>
04	05	
Read Response from Intent	Response Status Details	
<i>Handle the response data returned from the eMudhra app</i>	<i>Process different status values and extract transaction details</i>	

Step 1: Select or create PDF document as per the requirement

1. Client app should initiate the eSign v2 transaction using the API provided.

a. Endpoint: *esignexternal/v2_1/signdoc*

2. After the response is received from initiate API, using the *txnID* and *responseCode* from the *txnRef*.

a. *txnID*: Transaction ID sent in the initiate API request.

b. *responseCode*: *responseCode* received in the initiate API response.

Use the below method to call the eMudhra ARR UI via intent:

```
public void startNewActivity(String responseCode, String txnId) {
try {
    String txnRef = base64Encode((txnId + "|" +
responseCode).getBytes(StandardCharsets.UTF_8));
    Intent intent = new Intent(PreviewActivity.this, VerifyASPREqAndSign.class);
    intent.putExtra("txnRef", txnRef);
    intent.putExtra("aadhaarEncryptValue", encryptedValue);
    intent.putextra("envMode",envMode);
    startActivityForResult(intent, INTENT_REQUEST_CODE);
} catch (ActivityNotFoundException ex) {} }
```

Property Name	Data Type	Remarks
<i>txnRef</i>	<i>String</i>	<i>txnRef = Base64(transactionID + ' ' + responseCode)It is a base64 encoding string formed by concatenating transactionID and responseCode (separated by a pipe character).transactionID should be same as the transactionID sent in the Initiate eSign request API. ResponseCode should be same as the responseCode receive in the Initiate eSign response.</i>
<i>envMode</i>	<i>String</i>	<i>Environment mode selection: 1 -> Sandbox 2 -> Production</i>

Required Dependencies for Client App

```
dependencies {  
    implementation(libs.converter.simplexml)  
    {  
        exclude(group = "org.xmlpull", module = "xmlpull")  
        exclude(group = "xpp3", module = "xpp3")  
    }  
    api(libs.converter.gson)  
    implementation(libs.rootbeer)  
    implementation(libs.jacksonXml)  
    {  
        exclude(group = "xpp3", module = "xpp3")  
        exclude(group = "org.xmlpull", module = "xmlpull")  
        exclude(group = "javax.xml.stream", module = "stax-api")  
    }  
}
```

Step 2: Authentication Mode Selection

ASP can select the authentication mode either OTP or Biometric while creating the eSign request.

Based on the requested mode, the user will get OTP or Biometric option in the eMudhra App.

Biometric Mode

User enters Aadhaar number and selects biometric device from dropdown. Ensure RD Service app is installed.

12:49

Verify Request and Sign

We have received a request to Perform eSign using the Aadhaar OTP/Biometric.

ASP Name

eSign Retail Common ASP Linked Everywhere

ASP ID

MASP

Transaction ID

2025091512484510001878

Aadhaar Number / Virtual ID

(Enter your 12 digit Aadhaar Number / 16 digit Virtual ID (VID). OTP will be sent to the mobile linked with Aadhaar.)

Fingerprint Devices

Select Device

☐ I agree to the Consent taken in accordance to Aadhaar Regulations and Aadhaar Act.

Cancel

Perform eSign

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OTP Mode

User enters Aadhaar number and requests OTP for verification.

Verify Request and Sign

We have received an request to Perform eSign using the Aadhaar.

ASP NameeSign Retail Common ASP Linked Everywhere

ASP IDMASP

Transaction ID2022121617585110001090

Aadhaar Number / Virtual ID

(Enter your 12 digit Aadhaar Number / 16 digit Virtual ID (VID). OTP will be sent to the mobile linked with Aadhaar.)

OTP has been sent to mobile number *****8926

Aadhaar OTP

Resend OTP

(Please enter the OTP received to your entered Aadhaar linked Mobile Number.)

☐ I agree to the Consent taken in accordance to Aadhaar Regulations and Aadhaar Act.

Cancel

Perform eSign

In OTP mode, the user enters the OTP received on the registered mobile number before proceeding.

Verify Request and Sign

We have received an request to Perform eSign using the Aadhaar.

ASP NameeSign Retail Common ASP Linked Everywhere

ASP IDMASP

Transaction ID2022121617585110001090

Aadhaar Number / Virtual ID

(Enter your 12 digit Aadhaar Number / 16 digit Virtual ID (VID). OTP will be sent to the mobile linked with Aadhaar.)

OTP has been sent to mobile number *****8926

Aadhaar OTP

Resend OTP

(Please enter the OTP received to your entered Aadhaar linked Mobile Number.)

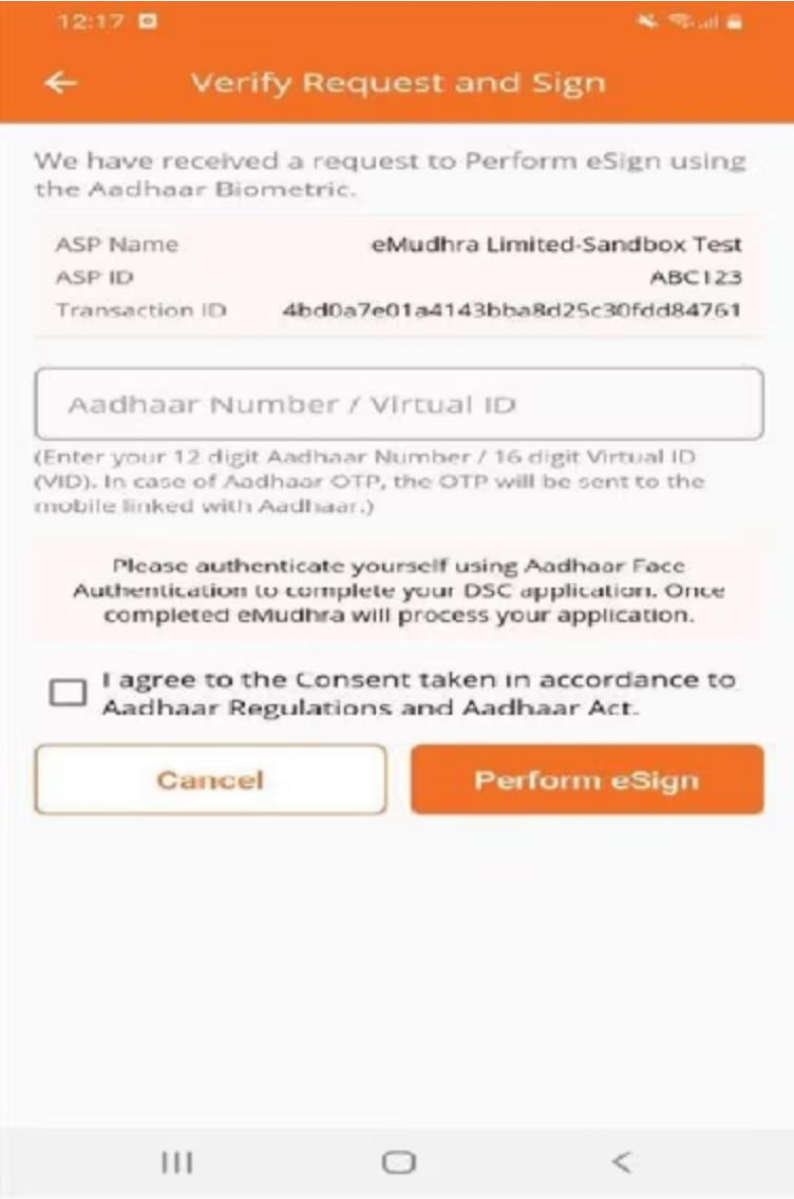
☐ I agree to the Consent taken in accordance to Aadhaar Regulations and Aadhaar Act.

Cancel

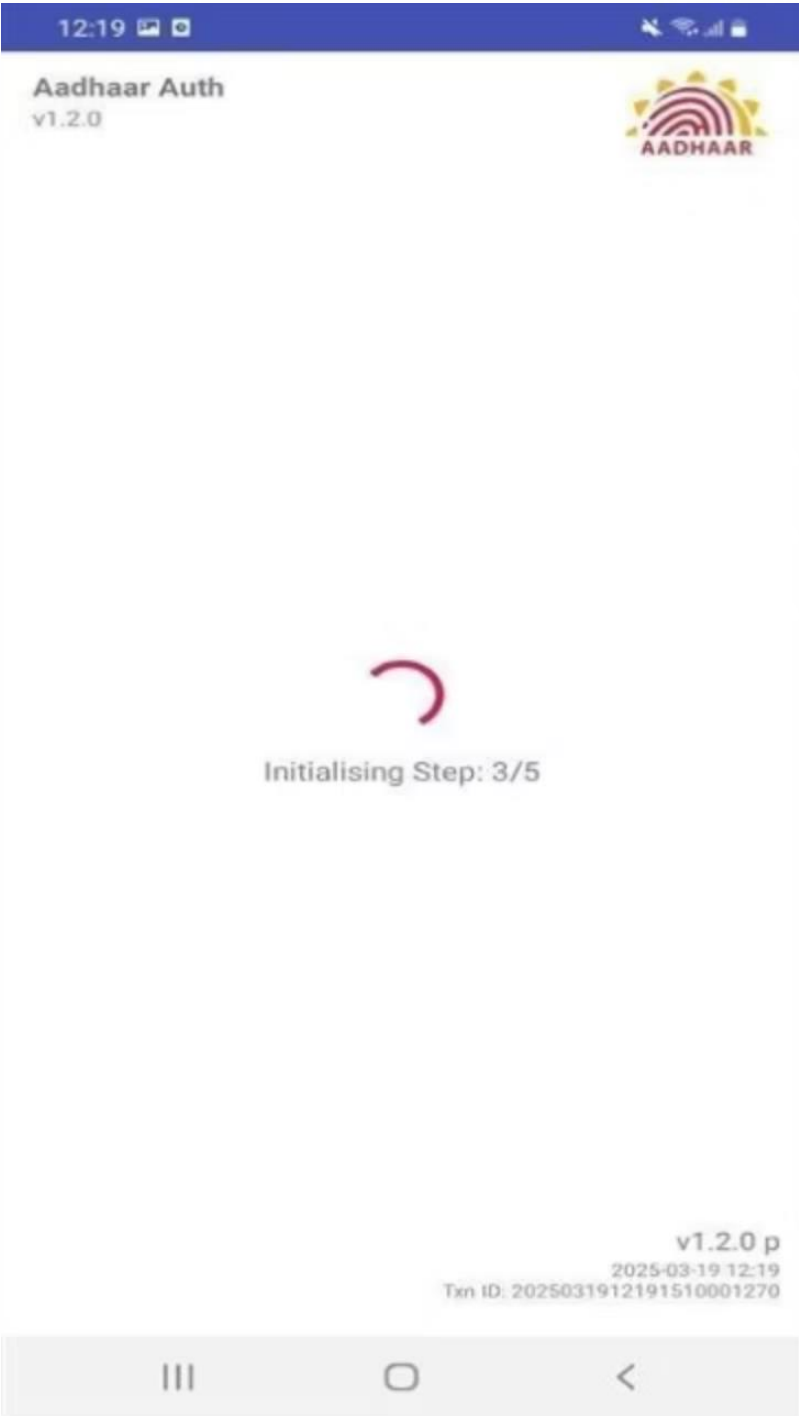
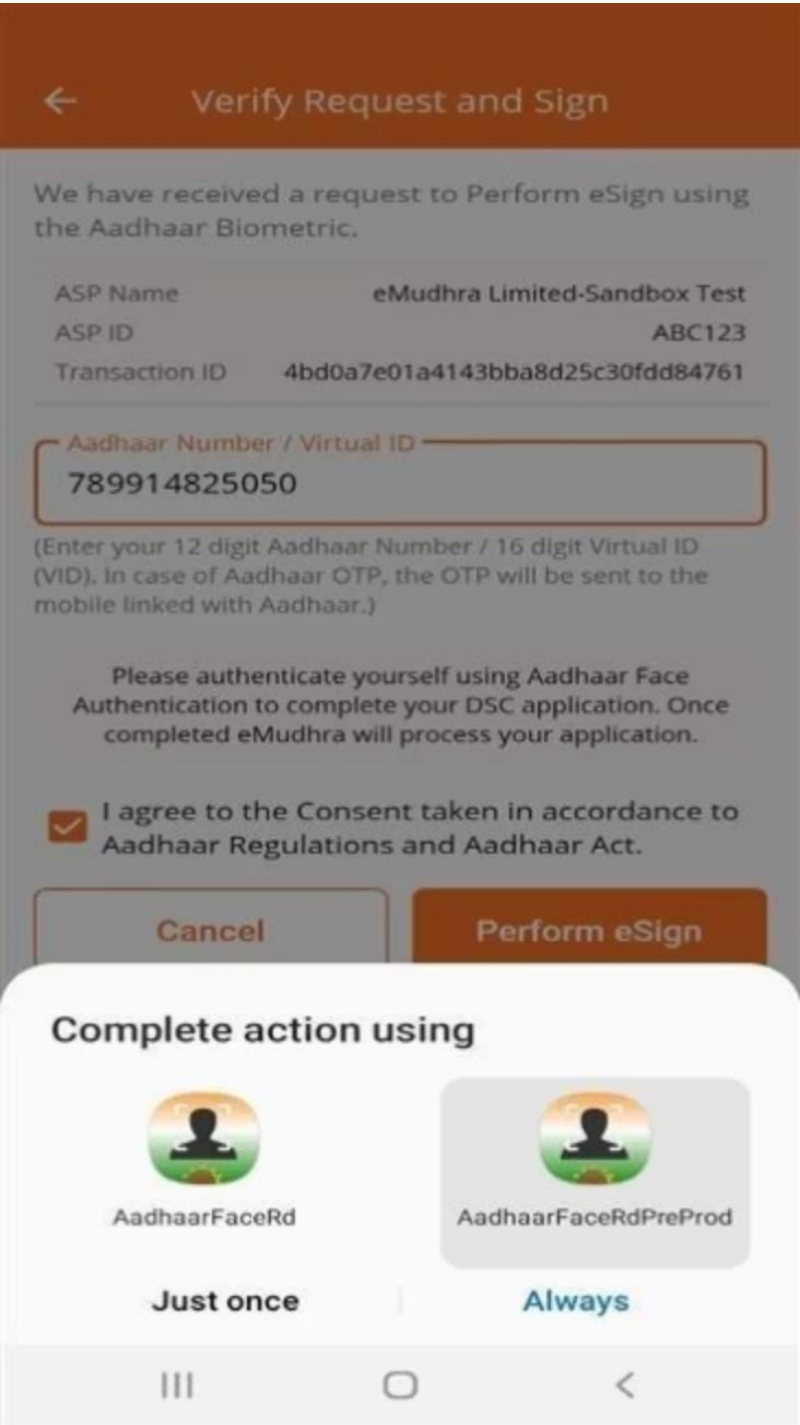
Perform eSign

Face Recognition

If AadhaarFaceRD app is installed, user can proceed with face authentication.



In case, if the user has the AadhaarFaceRD app, it will show the options to proceed when clicked on perform eSign, as shown below in the screenshot and will be redirected for face auth process.



Iris Recognition & Consent Process

Iris Recognition

User enters Aadhaar number and selects Iris device from dropdown. Ensure RD Service app is installed.

11:46

Verify Request and Sign

We have received a request to Perform eSign using the Aadhaar OTP/Biometric.

ASP Name

eSign Retail Common ASP Linked Everywhere

ASP ID

MASP

Transaction ID

202509151145591000215

Aadhaar Number / Virtual ID

(Enter your 12 digit Aadhaar Number / 16 digit Virtual ID (VID). OTP will be sent to the mobile linked with Aadhaar.)

Iris Devices

Select Device

☐ I agree to the Consent taken in accordance to Aadhaar Regulations and Aadhaar Act.

Cancel

Perform eSign

III

Step 3: Consent & perform eSign

User must select the checkbox to agree to the consent and click on 'Perform eSign'.

Verify Request and Sign

We have received an request to Perform eSign using the Aadhaar.

ASP Name

eSign Retail Common ASP Linked Everywhere

ASP ID

MASP

Transaction ID

202212161754171000194

Aadhaar Number / Virtual ID

(Enter your 12 digit Aadhaar Number / 16 digit Virtual ID (VID).
OTP will be sent to the mobile linked with Aadhaar.)

Biometric Devices

Evolute Identi5

☐ I agree to the Consent taken in accordance to
Aadhaar Regulations and Aadhaar Act.

Cancel

Perform eSign

Step 4 & 5: Response Handling

Step 4: Read Response from Intent

The response (status, errorMsg, responseXML) is received back in the intent result.

```
@Overrideprotected void onActivityResult(int requestCode, int resultCode, @Nullable Intent data) {  
    super.onActivityResult(requestCode, resultCode, data);  
    String responseXML, xmlValue, respPkcs7 = "", responseStatus = "";  
    if (requestCode == INTENT_REQUEST_CODE) {  
        if (data != null) {  
            responseXML = data.getStringExtra("responseXML");  
            String status = data.getStringExtra("status");  
            String errorMsg = data.getStringExtra("errorMsg");  
        }  
    }  
}
```

Step 5: Response Status Details

The meaning of different status values is as follows:

Status = 0

*responseXML is empty,
errorMsg has value.*

Status = 1

*responseXML has value,
errorMsg is empty.*

Status = 2

*responseXML has value,
errorMsg also has value.*

In case of status 1 & 2, the responseXML will contain the details and must be parsed to extract the exact transaction status and details.